

An app a day keeps the doctor away

Devices apart, there are several mobile apps and websites too that help people monitor their health and stay fit. Here are some examples:

- Beeminder (www.beeminder.com)
- Runkeeper (runkeeper.com)
- Retrofit (www.retrofitme.com)
- MapMyFitness (www.mapmyfitness.com)
- Cardio (www.cardio.com)
- Azumio (www.azumio.com)
- PocketYoga (www.pocketsports.com)
- DietBet (www.dietbet.com)

being successfully implemented and we have contributed to the reduction in maternal mortality and infant mortality rates. Women and children form over half of the patients who use the system. We have used the same infrastructure to educate pregnant women on what to expect when expecting," says Gupta.

Understanding rural India

ReMeDi is one of the oldest telemedicine solutions, made for India, in India. It has around 700 village centres, each one located at a population of 5000 or less, in the most rural areas. More than 200,000 consultations have been carried out in the last couple of years, where patients have actually paid for the consultations.

The ReMeDi medical data acquisition unit and ReMeDi software can measure basic physiological parameters of the patient and provide vital information to a doctor for preliminary diagnosis. Video conferencing integrated into this solution works even at a bandwidth as low as 32 kbps. With both the capital and operational expenses reduced, it is an extremely cost-effective solution for telemedicine applications. After a minimal amount of training, the solution can be used by anybody who is familiar with the usage of computers.

"ReMeDi facilitates the basic diagnosis of a patient remotely by a doctor by enabling the real-time measurement of the patient's ECG, temperature, oxygen saturation and blood pressure, as well as real-time transmission of auscultation sounds of the patient, captured using the electronic stethoscope.

A preliminary diagnosis of cardio-vascular problems can be carried out using stethoscope, ECG and BP measurement. Using high-resolution images taken from a normal Web camera, doctors can provide eye care, dermatology and ENT services," says Rajeev Kumar.

Multiple consultations can take place simultaneously, with the server performing the task of session management. All the medical data is stored in the electronic medical records of patients on the central server and is available on-demand. Both real-time as well as store-and-forward modes of data transfer are supported. The client-server architecture also enables various other entities (pharmacies, clinics, diagnostic labs, etc) to be part of the network. It allows associated services to be offered to the consumers through this network.

The solution features customised billing, customer feedback, offline audits, etc. It is rugged, designed for rural conditions, and is very easy to use. It is completely isolated from external power supply, and is suitable for any kind of IP connectivity. It has a low-bandwidth requirement (32 kbps) for real-time audio, video and medical data transfer. In all, it is designed perfectly for Indian usage conditions.

Providing healthcare where it matters the most

In areas like Kargil or remote villages in Andaman & Nicobar, where there are no communication links, providing healthcare would require not just technology but also infrastructure.

The Indian Space Research Organisation's (ISRO) telemedicine pilot project was started in 2001, to overcome precisely this problem—and they have done it quite successfully. The initiative involves several system providers including Infinium, Apollo, OTRI, BEL-VEPRO and TeleVital, and connects remote health centres with city-based super speciality hospitals

using INSAT satellites.

The telemedicine system consists of customised hardware and software at both the patient's and doctor's end, and diagnostic equipment such as ECG, X-ray and pathology microscope/camera at the patient's end. All this is connected using a very small aperture terminal (VSAT) system and controlled by ISRO's network hub station. A telemedicine system consisting of simple computers and communication systems conveys the medical images and other patient-related information to the specialist doctors—either in advance or on a real-time basis through the satellite link—in the form of digital data packets. These packets are received at the specialist centre, where the images and other information are reconstructed so that the specialist doctor can study the data, diagnose, interact with the patient and suggest appropriate treatment during a video conference with the patient.

Presently, ISRO's telemedicine network has enabled 306 remote centres and 16 mobile telemedicine units to be connected to 60 super-speciality hospitals in major cities. About 150,000 patients are being treated every year.

"It is a myth that establishing a telemedicine platform is expensive and a challenge. We have been able to demonstrate that the platform costs can be brought down to less than a dollar per consultation. The basic requirement for setting up the system is hardware, software and a telecommunication link. Also, there is a significant reduction in the prices of these basic requirements, making them well within the reach of most hospitals. The expenses may even be recovered by nominal charges to the patients, which would be much less than the physical travelling costs incurred," comments Gupta.

"The need of the hour is standardisation of telemedicine practices and communication hardware protocol in the country," he adds. ●

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